



**TAILORED AUTOMATION
& INTRALOGISTICS**

**The Corrugated Plant
of **the Future**
Starts Here**



TAILORED AUTOMATION
& INTRALOGISTICS



1.268

Projects

28

Years

26

Agreements

194

Installations

WHAT WE DO

Industrial machinery

Installation & commissioning

Robotics & palletizing

Project management

360° turnkey solution

Technical auditing

Industrial Intelligence (Ing_PRO)

Operational efficiency and reduced manual handling

Seamless integration into your existing plant flow

Lower error rates, higher throughput, improved safety

Single point of contact for the entire project lifecycle

Design, execution, training, and startup from A to Z

Independent diagnosis, free from manufacturer bias

AI applied to operations, maintenance, and quality

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INGETRANS

REEL HANDLING CART

Maximum efficiency, safety
and precision in paper reel
handling and transport.



**ENGINEERING
EXCELLENCE**



**SMART
AUTOMATION**



**INTRALOGISTICS
SOLUTIONS**



**PERFORMANCE
DRIVEN**



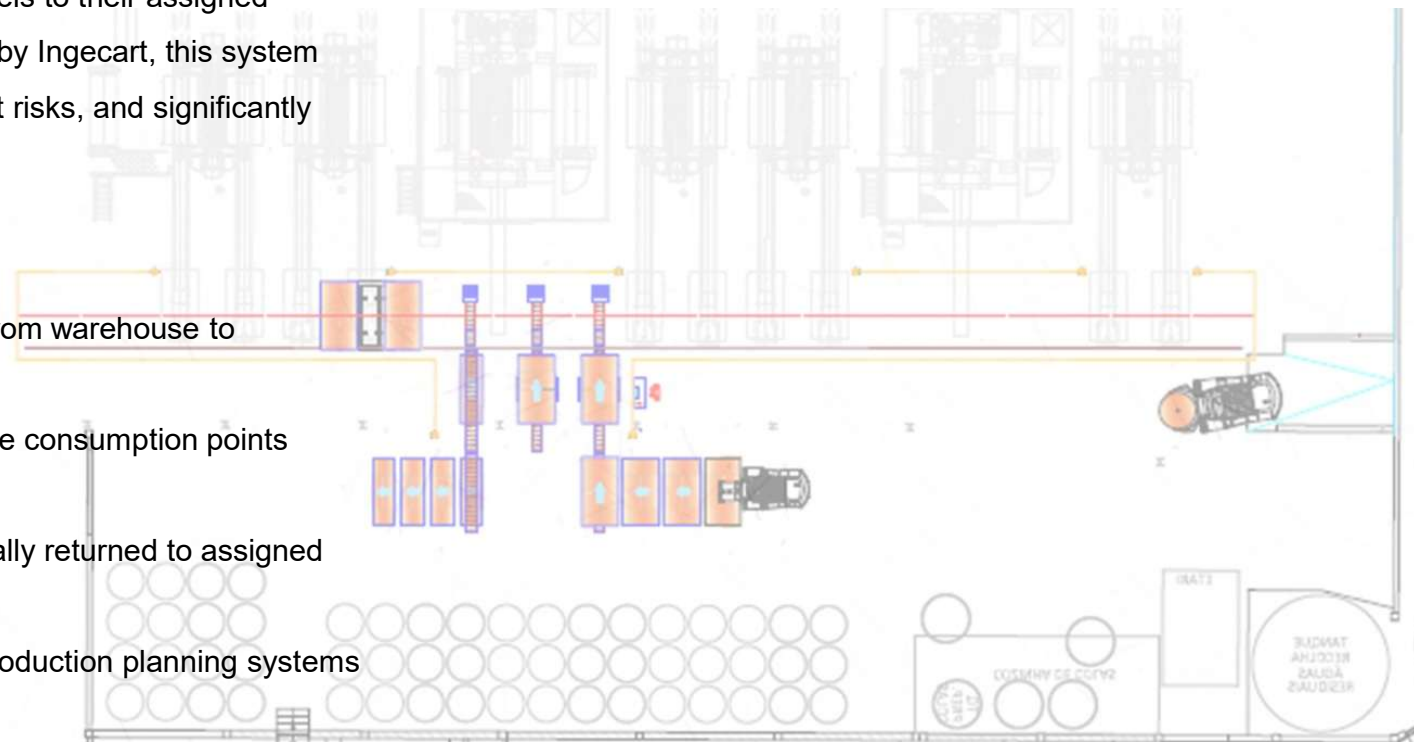
INTRALOGISTICS **INGETRANS**

Automated Corrugator Paper Reel Feeding System

INGETRANS is a complete automated solution for managing paper reel supply to multiple consumption points on the corrugator, including the automatic return of unused, or partially used, reels to their assigned storage locations. Designed and manufactured by Ingecart, this system eliminates forklift dependency, reduces accident risks, and significantly improves operational efficiency.

KEY FUNCTIONS

- Automated movement: Reels are transported from warehouse to corrugator without manual intervention
- Automated delivery: Precise delivery to multiple consumption points (single facers, roll stands)
- Automated return: Unused rolls are automatically returned to assigned warehouse locations
- Process integration: Works seamlessly with production planning systems



INTRALOGISTICS INGETRANS

Ingetrans– Corrugator Relel Change Cycle Time Calculation

Calculation basis: Delivery and return cycle for 5 rolls,
Minimum Order Length at 300 m/min

Description	Seconds	Minutes	Parameter	Value	Unit
Total cycle time – Delivery & return of 5 rolls	346.72	5.78	Corrugator speed	300	m/min
Average time per roll	69.34	1.16	Minimum order length at 300 m/min	1,733.6	meters

INTRALOGISTICS INGETRANS

TECHNICAL SPECIFICATIONS

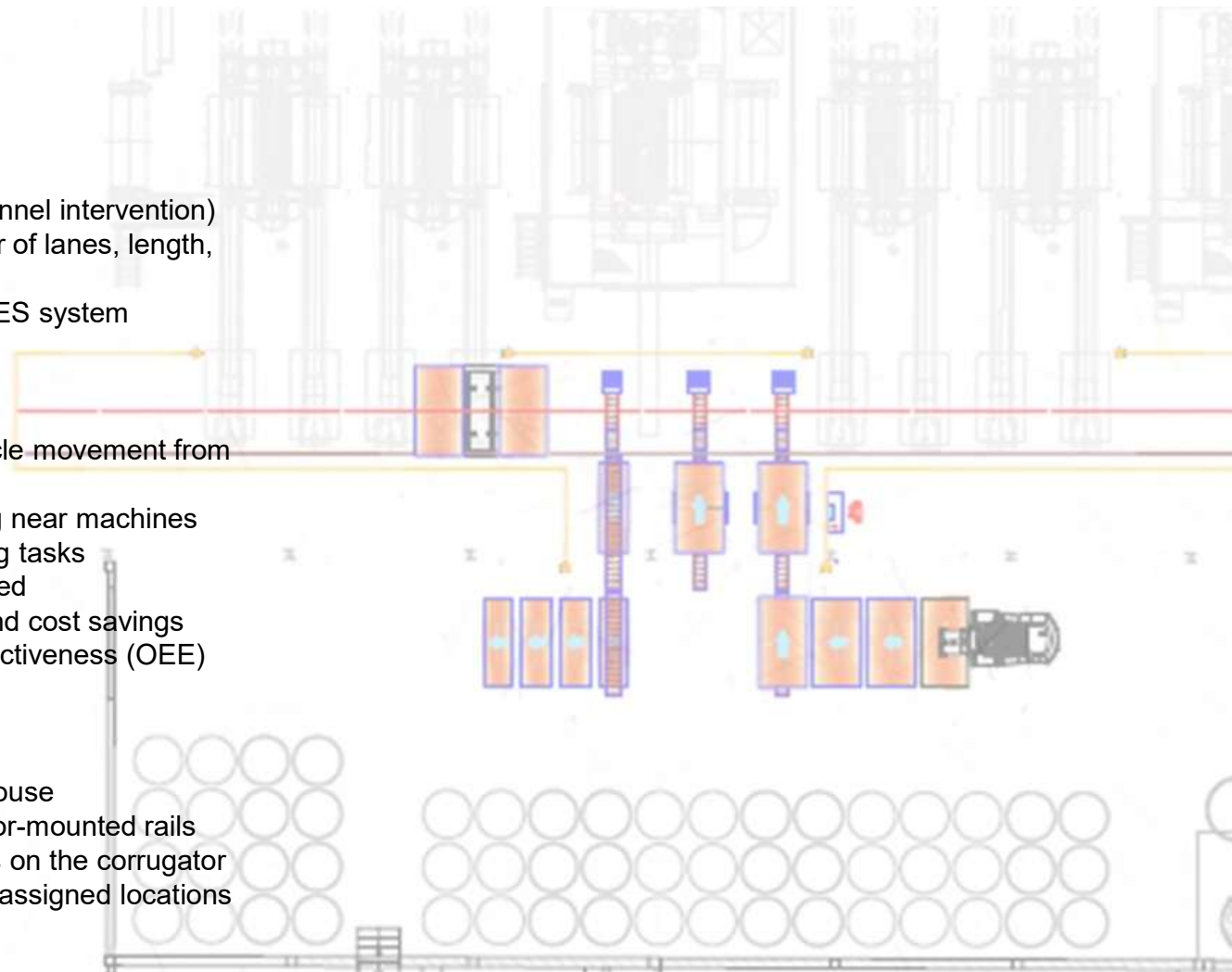
- System type: Surface rail-guided vehicle
- Track system: Floor-mounted rails
- Operation: Fully automated (no forklift or direct personnel intervention)
- Adaptability: Custom design per client layout (number of lanes, length, consumption points)
- Integration: Compatible with customer's planning / MES system

BENEFITS

- Eliminates forklift traffic – removes uncontrolled vehicle movement from corrugator work area
- Reduces accident risk – protects personnel operating near machines
- Labor cost reduction – minimizes manual roll handling tasks
- Minimizes downtime – rolls arrive exactly when needed
- Fast ROI – quick payback through efficiency gains and cost savings
- Increased profitability – higher overall equipment effectiveness (OEE)

HOW IT WORKS

1. Start – The process begins in the customer's warehouse
2. Transport – The Ingetrans carriage moves along floor-mounted rails
3. Delivery – Rolls are delivered to consumption points on the corrugator
4. Return – Unused rolls are automatically returned to assigned locations



More meters per minute. Zero chaos at the corrugator.

Safe productivity for corrugated plants.

More real line capacity: continuous, production-synchronized supply.

Automates reel movement, delivery, and return from warehouse to corrugator.

Creates safer workspaces by eliminating uncontrolled circulation.

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DRIVEN**

Plug&Play Easypack+Palletizer : Stack higher. Stack faster. Zero errors..

15 packages per minute.

🔌 No production stoppage. Plug, configure, and palletize.

🔧 Universal gripper. 50+ sizes. Zero changeover time.

No tools needed. Just a recipe.

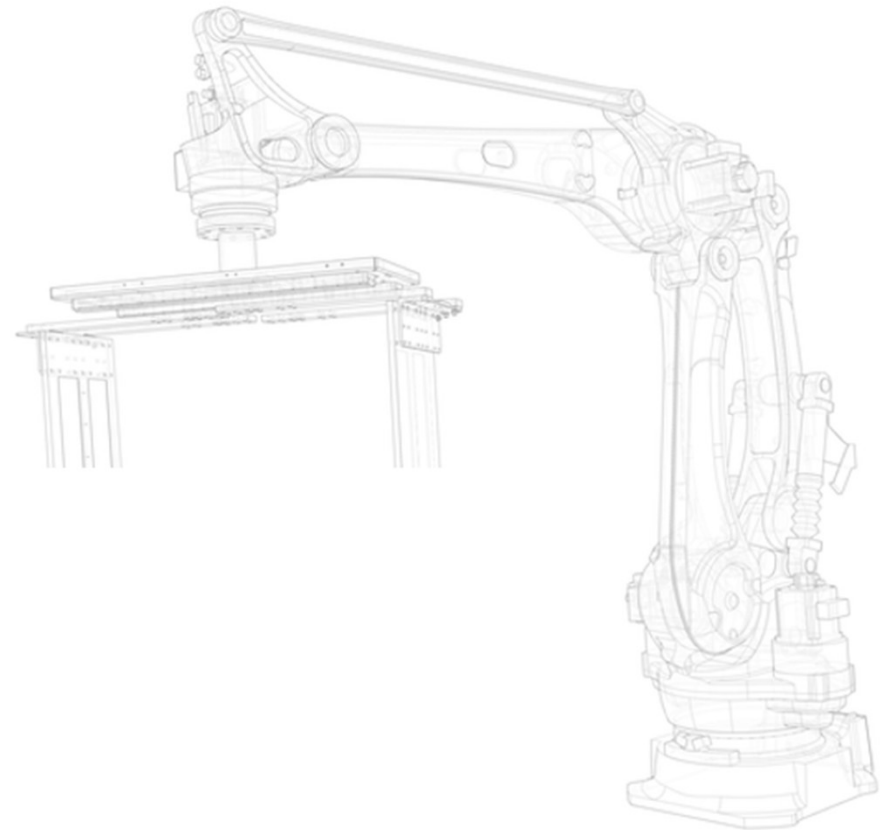
📊 ROI: Automation pays for itself.

🧠 Multi-language HMI + ERP. Real-time data.

Traceability and control from day one.

📦 Scale without redesigning your plant.

Growth doesn't have to be complicated.



Ingecart
ENGINEERING & AUDITING

LOGISTICS

GISTICS

by **KUKA**, Ingecart has develop
automated system for improving
management, efficiency
cost reduction .



Ingecart Intralogistics Management System

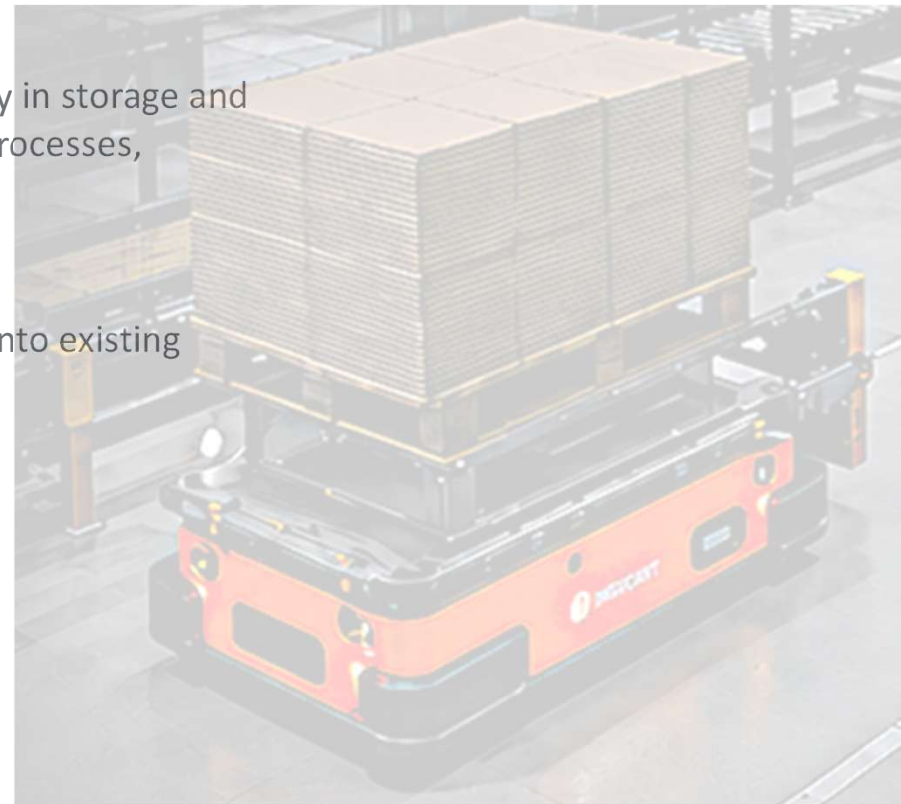
AMR-Based Solution

Ingecart Intralogistics Management System – AMR-Based Solution

Automate. Optimize. Maximize Every Square Meter.

Autonomous Mobile Robots (AMRs) significantly increase efficiency in storage and production environments. They automate transport and logistics processes, accelerating task execution and optimizing goods flow.

Ingecart's compact and flexible AMR models integrate seamlessly into existing supply chains – saving substantial time and resources.



Ingecart Intralogistics Management System

AMR-Based Solution

Fully automated movement of loads, WIP, and finished goods between warehouse, production, and converting machines

Real-time communication with production planning systems

Logistics Efficiency

Just-in-time delivery of materials to converting stations

Reduced waiting times and improved overall equipment effectiveness (OEE)

Synchronized material flow aligned with production pace

Maximum Space Utilization (WIP per m²)

Compact AMR design navigates narrow aisles and tight spaces

Optimized WIP placement in minimum footprint

Intelligent routing allows higher storage density without adding square meters

Flow Management Between WIP & Converting Machines

Dynamic management of work-in-progress buffers





SR-1400

WASTE SYSTEM

SR-1400

The complete waste collection
& transport system.

One system. All waste.

Casemakers | Die-cutters |
Corrugator | Sheet waste



LOW ENERGY



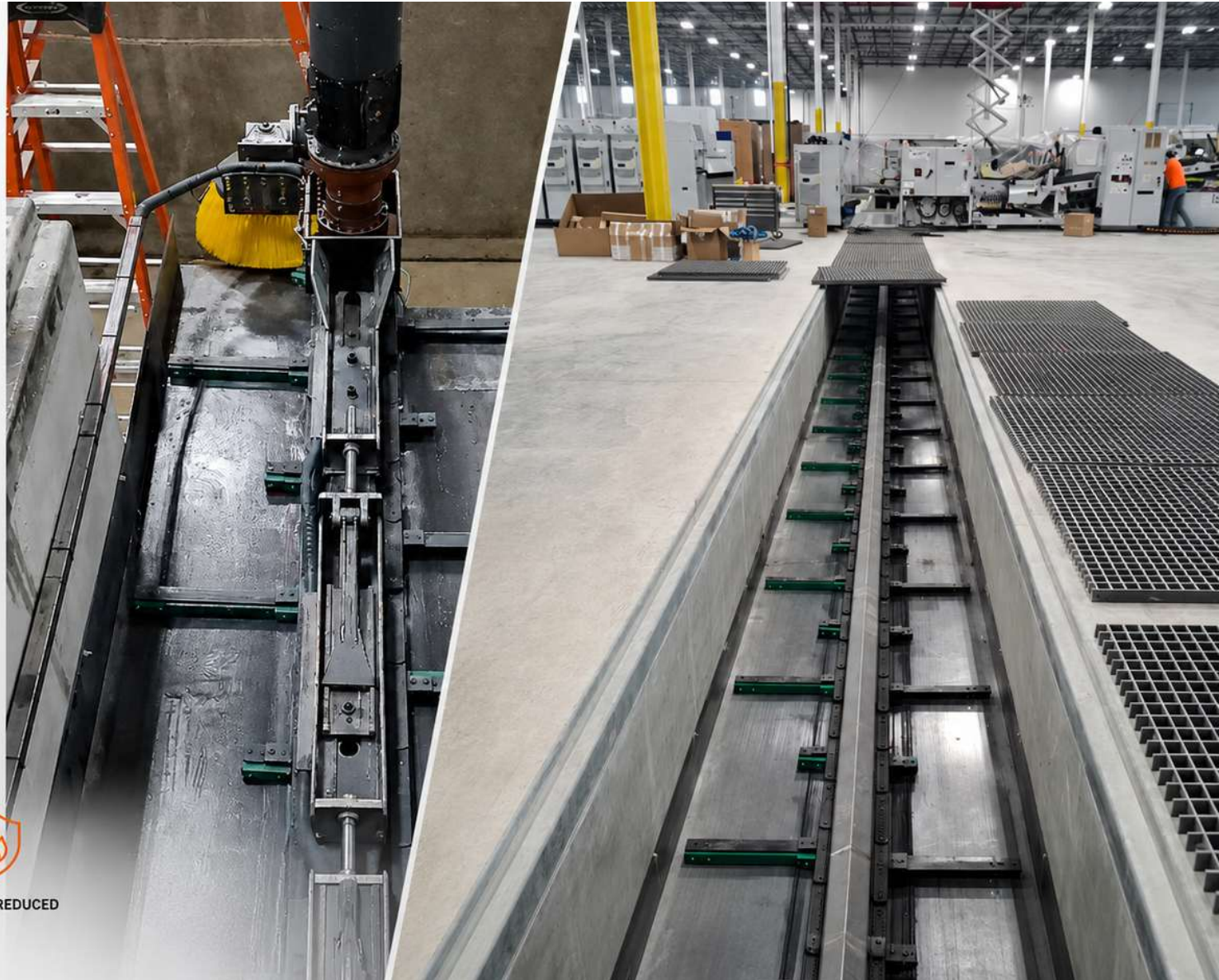
LOW NOISE



NO DUST



FIRE RISK REDUCED



SR-1400

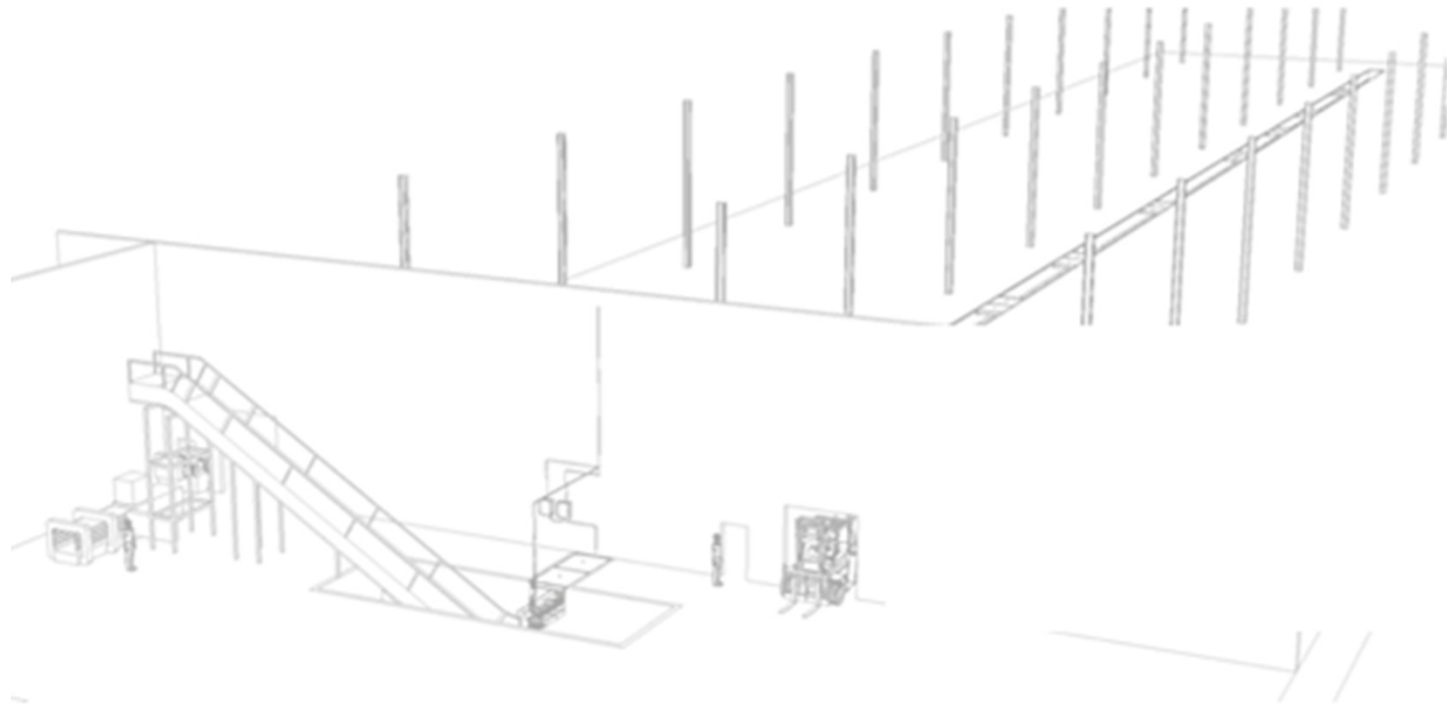
unified solution for collecting and transporting all waste

WHAT IT HANDLES

- Casemaker waste
- Die-cutter waste
- Corrugator waste
- Sheet waste

KEY BENEFITS

- Very low energy consumption
- Low noise level
- No airborne dust
- Reduced fire hazard



SR-1400

unified solution for collecting and transporting all waste

Auto-lubrication system

Sectional stop at each machine:

Safety push-buttons allow operators to stop non-operative sections, improving safety and saving energy.

Anti-wear plates at line ends

Special chain & arm design

Developed after studying multiple customer installations. Eliminates common fixing failures and constant maintenance issues.

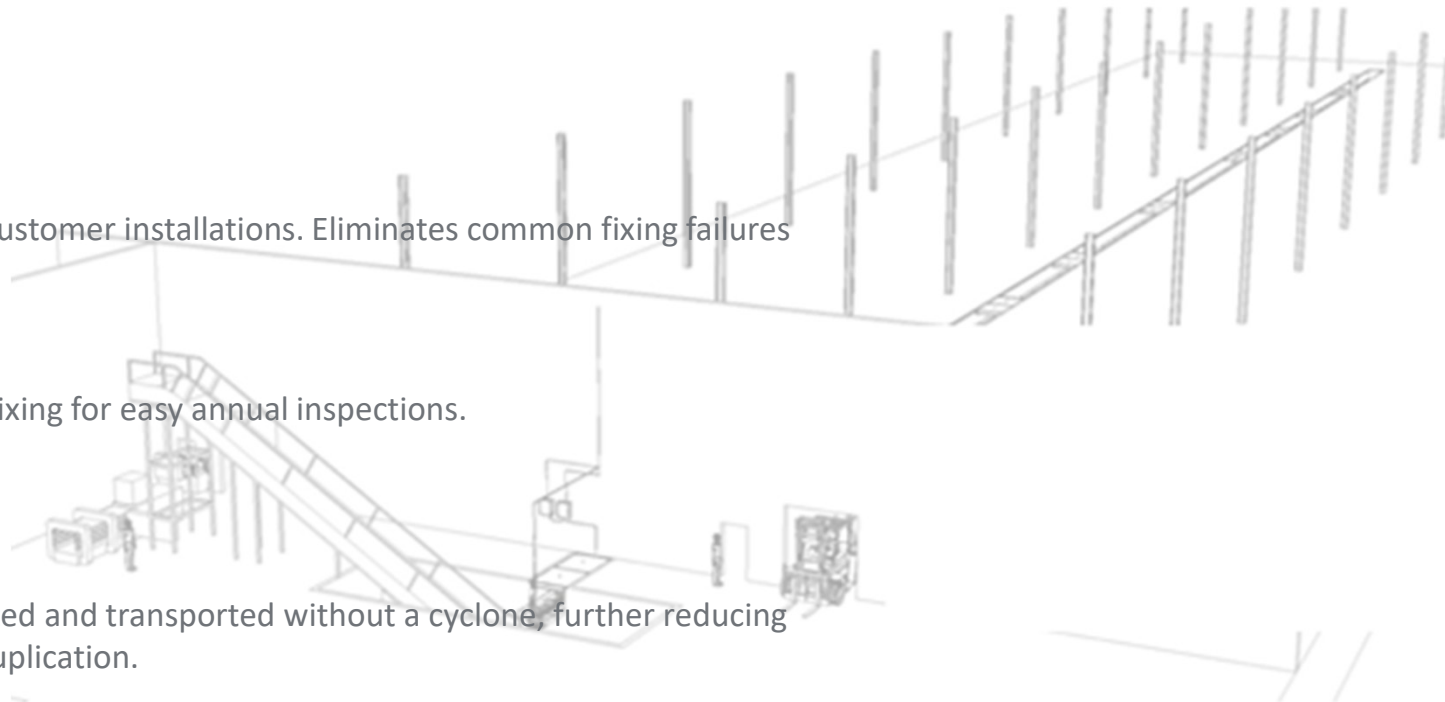
Chain link breaking strength

35,000 kg – Metal fins allow screw fixing for easy annual inspections.

Speed inverters for flow control

Complete system integration

Waste from the corrugator is collected and transported without a cyclone, further reducing energy costs and avoiding system duplication.





DIRECTION & AUDIT

Independent consulting for the **corrugated industry**

At Ingecart, we understand that corrugated plant managers often stand alone when making critical decisions. The complex technical reality requires advice from highly qualified specialists with no external interests.

Our mission is to be your journey partner. We present objective, impartial reality – analyzing different options together with their technical and organizational impacts.

WE SERVE YOU IN THREE MAIN SCENARIOS:

1 EXISTING FACTORIES



We analyze your equipment and processes. We present suggestions with clear justification. We help you evaluate offers for new or used equipment, assist with implementation, support staff training, and follow production improvements.

2 NEW PROJECTS



We help you develop financial viability plans and ROI calculations. Support includes budget development, engineering and technical studies, machinery purchase negotiation, risk management, training, and complete project follow-up from commissioning to new goal definition.



OBJECTIVE
ANALYSIS



INDEPENDENT
JUDGMENT



TECHNICAL &
ORGANIZATIONAL
IMPACT



IMPLEMENTATION
SUPPORT



TRAINING &
DEVELOPMENT



CONTINUOUS
IMPROVEMENT

CONSULTANCY SERVICES

Our consultancy service is a fundamental part of our offering. We specialize in expert advice and customized solutions to help optimize processes and achieve business objectives.

In our consultancy meetings, we adopt a collaborative approach:

- Listen carefully to concerns and objectives
- Examine existing processes and operations in detail
- Identify inefficiencies, bottlenecks, and improvement points
- Work closely with your team to analyze findings
- Discuss possible solutions, emerging technologies, and industry best practices
- Provide practical, actionable recommendations

WHY INGECART?



Complete independence from manufacturers and equipment vendors



Deep experience in paper, corrugated board, and production



Unbiased advice focused entirely on your success



End-to-end service: from initial audit to project completion





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